

EARTH FLYBY ANOMALIES FROM DIRECTION-DEPENDENT HQIV INERTIA SCREENING AND O-MAXWELL ORBITAL PROBES

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ABSTRACT. Background. Anderson et al. reported millimetre-per-second discrepancies between Doppler tracking and classical Newton+J₂ models for several Earth gravity assists. Among the four most quoted cases, two are clean (NEAR-Shoemaker 1998, Galileo I 1990), one is contaminated by thruster activity within an hour of closest approach (Cassini 1999), and one (Rosetta-I 2005) sits above the atmosphere with no agreed-upon classical explanation. We do not attempt to explain Cassini’s residual nor Galileo’s later flybys.

Approach. HQIV adds a direction-dependent inertia screen $f(a, \varphi) = a/(a + \varphi/6)$ with an angular-momentum and colatitude blend, a horizon metric channel split into isotropic and Lense–Thirring components with a derived release fraction $\lambda = \gamma \sin^2 \theta \rho_{\text{pol}}$ ($\gamma = 2/5$), a co-spinning lapse contribution in the horizon channel only, an effective gravitational coupling $G_{\text{eff}}/G_0 = (\varphi/\varphi_{\text{ref}})^\alpha$ that scales the total acceleration as a time-factor, and an O-Maxwell chart slot with the spin coupling suppressed inside the horizon shell. The implementation is in `scripts/hqiv_orbital_flyby_omaxwell.py`; the algebra spine is mirrored in `Hqiv/Physics/OrbitalFlybyScaffold.lean`.

Solver. Classical RK4 with corrected (1, 2, 2, 1)/6 weights, verified to fourth-order on a Kepler benchmark, with Sun and Moon positions advanced *per substage* from the encounter epoch. The HQIV minus classical excess is stable to $\lesssim 0.01$ mm/s across $\Delta t \in [0.5, 4]$ s.

Result. At the nominal paper coupling ($\kappa_{\text{vac}} = 50$, $\kappa_\varphi = 5$) the calculator delivers, for the four catalog Earth flybys (literature Doppler residual in parentheses): NEAR +16.34 mm/s (+13.46), Galileo I +4.15 mm/s (+3.92), Cassini +0.74 mm/s (−0.50, thruster-contaminated; excluded from goodness), Rosetta-I −3.49 mm/s (+1.80, unexplained). The derived λ is the dominant single change versus the legacy scalar lapse ($\lambda = 0.5$ fixed, NEAR +33.5 mm/s). Replacing the derived λ by zero (“no Lense–Thirring 3-vector”) gives NEAR −1.91 mm/s, confirming the L–T channel is the load-bearing piece. No PDG mass or fitted ephemeris is used; the proton anchor shell $m_\oplus = 4$ enters only the φ readout.

1. SCOPE AND CLAIM DISCIPLINE

This is a falsifiable model and a reproducible integrator, not a proof that Anderson anomalies are HQIV in origin. We make the following claim discipline explicit:

- (1) **Locked equations.** A single LaTeX equation sheet (this manuscript, `scripts/hqiv_flyby_equations.py` and the Lean scaffold) is the only source of truth. CLI: `--equations latex` prints the same sheet used here.
- (2) **Per-mission honesty.** We only score the integrator against missions whose literature residual is plausibly clean (NEAR, Galileo I). Cassini’s −0.5 mm/s is well below its thruster injection budget and is reported but not scored. Rosetta-I’s sign mismatch is flagged as an open item. Galileo’s 1992 flyby (GLI-2) at ~ 300 km altitude is atmospheric-drag dominated and is not included.
- (3) **No PDG injection.** The horizon shell ladder $\varphi(m) = 2(m + 1)$ uses $m_\oplus = 4$ as a discrete index, not a measured mass.
- (4) **Comparisons use HQIV–classical.** Absolute classical Δv from the present RK4+J₂ baseline can reach m/s because of asymptotic-shell truncation; the relevant observable is the same-integrator difference.

2. MISSION ACCOUNTING

NEAR-Shoemaker (1998-01-23, $h_p = 539$ km). The largest reported residual (+13.46 mm/s) and the cleanest tracking. Anderson’s analysis already includes a conservative atmospheric drag estimate, but the residual sits above the drag uncertainty band (~ 1 mm/s for the ~ 539 km perigee). We treat the +13.46 mm/s as the target observation with $\mathcal{O}(1-2)$ mm/s systematic from atmospheric drag and orbit-determination geometry.

Galileo I (1990-12-08, $h_p = 960$ km). First-flyby residual +3.92 mm/s. At 960 km the atmospheric drag contribution is small but non-zero. Galileo’s second Earth flyby (1992) had a ~ 300 km perigee that placed it deep in the residual atmosphere; the corresponding Doppler residual was either consistent with zero or atmospheric-drag dominated. We score only GLI-1 against the model.

Cassini (1999-08-18, $h_p = 1175$ km). Anderson’s table reports -0.50 mm/s with notably larger error bars than NEAR or Galileo, because Cassini fired thrusters within roughly one hour of closest approach. The Doppler residual at that level is dominated by thruster Δv injection and is not a clean gravitational target. We *report* the HQIV value for transparency but *exclude* Cassini from the goodness-of-fit accounting.

Rosetta-I (2005-03-04, $h_p \approx 1955$ km). Anderson reports +1.80 mm/s. The ~ 1955 km perigee is well above any significant atmosphere, so drag is not the explanation for either the literature value or our sign mismatch. The model’s -3.49 mm/s prediction is an honest open item: either the geometry implementation (impact-plane azimuth, declinations) needs refinement against the actual ephemeris, or the HQIV horizon channel acquires a sign flip at low $|L_z|$ that is not yet captured by the derived λ . Listed in §10.

3. CLASSICAL BASELINE

Earth-centred inertial frame with spin $\boldsymbol{\omega} = \Omega_\oplus \hat{\mathbf{z}}$. The classical acceleration on the probe is

$$(1) \quad \mathbf{a}_{\text{cls}}(\mathbf{r}, t) = \mathbf{a}_N + \mathbf{a}_{J_2} + \mathbf{a}_{3b},$$

$$(2) \quad \mathbf{a}_N = -\frac{GM_\oplus}{r^3} \mathbf{r},$$

$$(3) \quad \mathbf{a}_{J_2} = -\frac{3}{2} \frac{J_2 GM_\oplus R_\oplus^2}{r^5} \begin{pmatrix} x(5z^2/r^2 - 1) \\ y(5z^2/r^2 - 1) \\ z(5z^2/r^2 - 3) \end{pmatrix},$$

$$(4) \quad \mathbf{a}_{3b}(\mathbf{r}, t) = \sum_{B \in \{\odot, \text{Moon}\}} GM_B \left(\frac{\mathbf{R}_B(t) - \mathbf{r}}{|\mathbf{R}_B(t) - \mathbf{r}|^3} - \frac{\mathbf{R}_B(t)}{|\mathbf{R}_B(t)|^3} \right),$$

with $J_2 = 1.08263 \times 10^{-3}$, $GM_\oplus = 3.986004418 \times 10^{14} \text{ m}^3\text{s}^{-2}$, and $R_\oplus = 6.378137 \times 10^6 \text{ m}$. The Sun and Moon positions $\mathbf{R}_B(t)$ use a low-precision mean-element approximation seeded from the encounter ISO date and advanced by t during the integration (§5).

Hyperbolic seed. For impact parameter b and asymptotic speed v_∞ ,

$$|\mathbf{h}| = |\mathbf{r} \times \mathbf{v}| = b v_\infty, \quad \frac{1}{2} v^2 - GM_\oplus/r = \frac{1}{2} v_\infty^2.$$

We seed at $r_{\text{start}} = 80 R_\oplus$ with the inbound velocity declination set to the mission’s δ_{in} and the impact-plane azimuth tuned so that a pure Kepler outbound declination matches the mission’s δ_{out} .

Asymptotic Doppler readout. We sample on the outbound at $r > 40 R_\oplus$ and reduce to v_∞ via

$$v_{\infty, \text{shell}} = \sqrt{v^2 - 2GM_\oplus/r}, \quad \Delta v = \langle v_{\infty, \text{shell}} \rangle_{\text{out}} - \langle v_{\infty, \text{shell}} \rangle_{\text{in}}.$$

The averages are over all post-CA (post-encounter) and pre-CA samples in the shell, so the diagnostic is invariant to the precise truncation step.

4. HQIV LAYER

4.1. Shell ladder, lapse contributions, and modified inertia. The Lean-proved invariants $\alpha = 3/5$, $\gamma = 2/5$, $\varphi(m) = 2(m+1)$ (`alpha.eq.3.5`, `phi.shell_def`) fix the algebra. Earth's anchor shell is $m_{\oplus} = 4$, $\varphi_{\text{ref}} = 10$. The homogeneous scale is $\varphi_{\text{hom}} \approx 2cH_0$, and the local readout

$$(5) \quad \varphi_{\text{loc}}(\mathbf{r}) = \varphi_{\text{hom}} \frac{\varphi(m_{\oplus})}{1 + r/R_{\oplus}}.$$

The flyby calculator carries three lapse contributions:

$$(6) \quad \epsilon_{\text{spin}}(r) = \frac{2\Omega_{\oplus}R_{\oplus}}{c} \left(\frac{R_{\oplus}}{r} \right)^2 \times \sin^2 \theta_{\text{geo}},$$

$$(7) \quad \epsilon_{\text{yr}}(t) = \frac{v_{\oplus}}{c} \hat{\mathbf{v}} \cdot \hat{\mathbf{v}}_{\oplus}(t),$$

$$(8) \quad \epsilon_{\text{gal}}(t) = \frac{2(v_c/c) f_{\text{disk}}}{D_R} \frac{v_{\oplus}}{v_c} \hat{\mathbf{v}}_{\oplus}(t) \cdot \hat{\mathbf{v}}_{\text{gal}},$$

where the colatitude weight $\sin^2 \theta_{\text{geo}}$ comes from the requirement that a co-spinning horizon drag must vanish on the spin axis, $f_{\text{disk}} = M_{\text{disk}}(< R_0)/(v_c^2 R_0/G)$ measures the disk-supported fraction of the Sun's circular speed, and $D_R = 1 + \frac{\gamma}{2}(c/v_c)^2$ is the Rindler-horizon denominator from the galactic angular-momentum lapse. In the paper run the annual term is not used ($\lambda_{\text{yr}} = 0$) and the galactic term contributes at the 10^{-10} level.

Two effective φ readouts are then defined:

$$(9) \quad \varphi_{\text{eff,full}} = \varphi_{\text{loc}} + 6a_{\text{loc}} \max(0, \epsilon_{\text{spin}} + \epsilon_{\text{gal}} + \lambda_{\text{yr}}\epsilon_{\text{yr}}),$$

$$(10) \quad \varphi_{\text{eff,part}} = \varphi_{\text{loc}}.$$

$\varphi_{\text{eff,part}}$ feeds the particle geodesic; $\varphi_{\text{eff,full}}$ feeds the horizon metric channel. The repartition prevents the lapse contributions from *double-counting* as both a geodesic divide and a metric-side acceleration.

4.2. Modified inertia geodesic. The Brodie/main.tex particle action $S_{\text{particle}} = -m_g \int f(a, \varphi) ds$ yields the modified geodesic

$$(11) \quad m_i \mathbf{a} = m_g \mathbf{a}_{\text{GR}}, \quad m_i = m_g f_{\text{part}}, \quad \mathbf{a} = \mathbf{a}_{\text{GR}}/f_{\text{part}},$$

with $f_{\text{part}} = f(a_{\text{probe}}, \varphi_{\text{eff,part}})$ and a_{probe} the unscaled magnitude of $\mathbf{a}_{\text{N}} + \mathbf{a}_{J_2} + \mathbf{a}_{3b}$.

4.3. Direction-dependent screen weight.

$$(12) \quad a_{\text{eq}} = |g| + \frac{h^2}{r^3} \left(\frac{h_z}{h} \right)^2, \quad h_z = |(\mathbf{r} \times \mathbf{v})_z|,$$

$$(13) \quad \rho_{\text{pol}} = 1 - (h_z/h_{\text{ref}})^2, \quad h_{\text{ref}} = b v_{\infty},$$

$$(14) \quad h_{z,\text{eff,asym}}^2 = h_z^2 + \frac{h_{\text{ref}}^2}{(m+1)^2}, \quad h_{z,\text{eff,loc}}^2 = h_z^2 + \frac{h^2}{(m+1)^2},$$

$$(15) \quad \varphi_{\text{pol}} = \varphi_{\text{eff,full}} \left(1 + \rho_{\text{pol}} \max\left(\left[\frac{h_{\text{ref}}^2}{h_{z,\text{eff,asym}}^2} - 1 \right]_+, \left[\frac{h^2}{h_{z,\text{eff,loc}}^2} - 1 \right]_+ \right) \right),$$

$$(16) \quad f_{\text{full}} = \sin^2 \theta f(a_{\text{eq}}, \varphi_{\text{eff,full}}) + \cos^2 \theta f(|g|, \varphi_{\text{pol}}), \quad \sin^2 \theta = 1 - (z/r)^2,$$

$$(17) \quad w = 1 - f_{\text{full}}.$$

The shell-ladder floors $h_{z,\text{eff}}^2 = h_z^2 + (\cdot)/(m+1)^2$ replace the older ad-hoc $\max(h_z, 0.01h_{\text{ref}})$ regulariser with a physically motivated discrete contribution from the shell index, so the polar-fiber φ boost saturates at the lattice level rather than at a chosen 10^{-2} floor.

4.4. Horizon metric channel (repartitioned). The acceleration deficit between the full and particle screens is delivered as a metric-side force, not via further geodesic division:

$$(18) \quad \mathbf{a}_{\text{hor}} = \frac{\mathbf{a}_{\text{GR}}}{f_{\text{part}}} \left(\frac{f_{\text{part}}}{f_{\text{full}}} - 1 \right),$$

split into an isotropic radial component and a Lense–Thirring-style tangential component along $\hat{\boldsymbol{\omega}} \times \hat{\mathbf{r}}$:

$$(19) \quad \mathbf{a}_{\text{hor}} = (1 - \lambda) \mathbf{a}_{\text{hor}}^{\text{iso}} + \lambda |\mathbf{a}_{\text{hor}}| \widehat{\boldsymbol{\omega} \times \mathbf{r}}, \quad \lambda = \gamma \sin^2 \theta \rho_{\text{pol}}.$$

The fraction λ is *derived*, not fitted: $\gamma = 2/5$ is the locked HQIV constant, $\sin^2 \theta$ vanishes on the spin axis (no horizon drag at the pole), and $\rho_{\text{pol}} = 1 - (h_z/h_{\text{ref}})^2$ vanishes for equatorially locked orbits (no L–T release when $|L_z| = |L|$). Replacing λ by the previous fixed value 0.5 overshoots NEAR by $\sim 2.5\times$; setting $\lambda = 0$ drives NEAR’s HQIV excess negative (§8).

4.5. Vacuum chart slot.

$$(20) \quad \mathbf{g}_{\text{vac}} = -\frac{\gamma}{6} (\varphi \nabla \dot{\theta}' + \dot{\theta}' \nabla \varphi), \quad \mathbf{a}_{\varphi} = \frac{\alpha}{4\pi} \ln(\varphi + 1) \nabla \varphi,$$

$$(21) \quad \mathbf{a}_{\text{HQIV}} = w(1 - \beta^2) (\kappa_{\text{vac}} \mathbf{g}_{\text{vac}} + \kappa_{\varphi} \mathbf{a}_{\varphi}), \quad \beta = |\mathbf{v}|/c.$$

The signed spin–phase tip $\dot{\theta}'$ is suppressed inside the horizon shell (`suppress_vacuum_spin_coupling=True`); the spin-odd dependence at flyby altitudes is now carried by the horizon channel (18) rather than duplicated in the vacuum source.

4.6. Effective Newton coupling and time-factor application.

$$(22) \quad \frac{G_{\text{eff}}}{G_0} = 1 + \left((\varphi/\varphi_{\text{ref}})^{\alpha} - 1 \right) w.$$

G_{eff} rescales coordinate time, not individual gravitational sources: the total acceleration vector is multiplied by G_{eff}/G_0 at the end of the per-step assembly, after the geodesic divide, the horizon channel, and the vacuum slot (`geff_as_time_factor=True`). Applying it instead to each GM term separately is a double-count of the screen and inflates the excess at low w .

4.7. Putting it together. Per integration step at $(\mathbf{r}, \mathbf{v}, t)$,

$$(23) \quad \mathbf{a}(\mathbf{r}, \mathbf{v}, t) = \frac{G_{\text{eff}}}{G_0} \left[\frac{\mathbf{a}_{\text{cls}}(\mathbf{r}, t)}{f_{\text{part}}} + \mathbf{a}_{\text{hor}}(\mathbf{r}, \mathbf{v}) + \mathbf{a}_{\text{HQIV}}(\mathbf{r}, \mathbf{v}) \right],$$

with $\mathbf{a}_{\text{cls}}$ from §3 evaluated at the substage time and $\mathbf{a}_{\text{hor}}$, $\mathbf{a}_{\text{HQIV}}$ as above.

5. NUMERICAL INTEGRATOR

5.1. RK4 (corrected weights). For state $\mathbf{y} = (\mathbf{r}, \mathbf{v})$ with $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}, t) = (\mathbf{v}, \mathbf{a}(\mathbf{r}, \mathbf{v}, t))$ and step Δt ,

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(\mathbf{y}_n, t_n), \\ \mathbf{k}_2 &= \mathbf{f}(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_1, t_n + \frac{\Delta t}{2}), \\ \mathbf{k}_3 &= \mathbf{f}(\mathbf{y}_n + \frac{\Delta t}{2} \mathbf{k}_2, t_n + \frac{\Delta t}{2}), \\ \mathbf{k}_4 &= \mathbf{f}(\mathbf{y}_n + \Delta t \mathbf{k}_3, t_n + \Delta t), \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned}$$

The repository previously contained a transcription bug ($\frac{1}{6}(k_1 + 2k_2 + k_3 + 2k_4)$) that produced 50% error after 1000 steps on a damped harmonic-oscillator benchmark; the corrected (1, 2, 2, 1) weights reach 5×10^{-5} at the same step count. A unit test (`test_rk4_fourth_order_convergence_on_kepler`) now requires the energy-error ratio at $\Delta t = 120$ vs 60 to exceed 12 on a circular Kepler orbit at $3R_{\oplus}$.

5.2. Time-advancing ephemeris. $\mathbf{a}_{\text{cls}}$ contains the Sun/Moon differential tide. Both bodies move appreciably during a multi-hour flyby (Moon $\sim 0.5^\circ/\text{h}$). The integrator passes the substage time $t_n + c_i \Delta t$ into the ephemeris call and snaps it to 60-s buckets so the LRU cache hits across the four RK4 substages without freezing the ephemeris. A unit test (`test_sun_moon_advance_with_time_offset`) verifies the Sun moves $\sim 1^\circ$ and Moon $\sim 13^\circ$ per simulated day.

5.3. Step-size policy. For Earth flybys the calculator defaults to $\Delta t = 4$ s for $v_\infty > 12$ km/s and $\Delta t = 2$ s otherwise, with $r_{\text{escape}} = 40 R_{\text{lapse}}$, $t_{\text{max}} = 2.5 \times 10^5$ s. The HQIV minus classical excess is dt-converged to ~ 0.01 mm/s for NEAR across $\Delta t \in [0.5, 4]$ s:

Δt [s]	Δv_{cls} [mm/s]	Δv_{HQIV} [mm/s]	excess [mm/s]
4.0	−928.158	−911.815	+16.3425
2.0	−928.155	−911.812	+16.3425
1.0	−928.132	−911.790	+16.3425
0.5	−928.123	−911.771	+16.3523

Absolute $|\Delta v|$ drifts by ~ 0.04 mm/s with Δt because of asymptotic-shell truncation rather than RK4 truncation; the same drift appears in the classical baseline and cancels in the difference.

5.4. HQIV channel evaluation at substages. Every HQIV term in (11), (18), (19), (22) is re-evaluated at each of the four substages with the substage $(\mathbf{r}, \mathbf{v}, t)$. In particular the L–T direction $\hat{\omega} \times \hat{\mathbf{r}}$, the derived λ , the colatitude factor $\sin^2 \theta$, the third-body tide, and the screened G_{eff} are all updated four times per macro-step. There is no operator-splitting and no fractional-step delay between the gravity assembly, the screen, and the metric channel.

6. REPRODUCIBILITY AND LEAN SPINE

Code. `scripts/hqiv_orbital_flyby_omaxwell.py` (calculator + CLI), `scripts/hqiv_flyby_equations.py` (locked equations), and `scripts/hqiv_flyby_gap_report.py` (Anderson-style comparison tables).

Tests. `scripts/test_hqiv_orbital_flyby_omaxwell.py` contains the unit suite (26 tests at the time of writing): geodesic divide isolation, repartition consistency, co-spin lapse zeros on the spin axis, L–T direction orthogonality, G_{eff} scalar behaviour, derived- λ vs equatorial-lock limit, RK4 4th-order convergence, ephemeris time advance, third-body tide vanishing at the geocentre.

Lean. `Hqiv/Physics/OrbitalFlybyScaffold.lean` defines `hqivFluidInertiaFactor`, `hqivFlybyScreenWeight`, `hqivScreenedGeffRatio`, and the anchor-shell limit lemmas. It does not yet contain the orbit-level λ derivation; that is listed in §10.

Commands.

```
python3 scripts/hqiv_orbital_flyby_omaxwell.py --paper-table \
    > papers/flyby_paper_table.tex
python3 scripts/hqiv_orbital_flyby_omaxwell.py --catalog all --run-all \
    --paper-nominal --output scripts/artifacts/orbital_flyby_results_paper.json
python3 scripts/hqiv_flyby_gap_report.py          # nominal + ablation tables
python3 -m unittest scripts.test_hqiv_orbital_flyby_omaxwell -q
```

7. NOMINAL PAPER COUPLING

Parameter	Symbol	Value
Vacuum chart scale	κ_{vac}	50
Metric- φ scale	κ_{φ}	5
Inertia screen	$f(a, \varphi)$	on
Modified geodesic $\mathbf{a}_{\text{GR}}/f_{\text{part}}$	—	on
Co-spinning lapse ϵ_{spin} (horizon only)	—	on
Galactic disk Rindler lapse ϵ_{gal}	—	on (tiny)
Annual lapse ϵ_{yr}	$\lambda_{\text{yr}} = 0$	off
L_z / colatitude blend	—	on
Horizon repartition	particle / metric	on
Horizon metric channel	$\mathbf{a}_{\text{hor}}$	on
Lense–Thirring vector	λ derived	on
G_{eff} as time-factor	—	on
Spin coupling in $\mathbf{g}_{\text{vac}}$	—	suppressed
Velocity screen $1 - \beta^2$	—	on

8. CATALOG SNAPSHOT (CORRECTED RK4, DERIVED λ)

Earth-centred runs, RK4 with (1, 2, 2, 1)/6 weights and time-advancing Sun/Moon.

Case	Lit. [mm/s]	r_{CA} [km]	$\langle 1-f \rangle_{\text{out}}$	Δv_{cls}	HQIV–cls
equator_to_pole	—	5775	1.98e-08	7170.37	6.43
equator_to_equator	—	5781	1.16e-09	1142.40	-0.01
near_1998	13.46	6899	7.46e-09	-928.16	16.34
galileo_1990	3.92	7328	1.25e-09	-294.33	4.15
cassini_1999	-0.50	7554	1.25e-09	-86.07	0.74
rosetta_2005	1.80	8302	1.25e-09	-1403.76	-3.49
generic_deep	—	11530	1.43e-09	-877.08	-0.00

Reading the table. r_{CA} is closest-approach radius; $\langle 1-f \rangle_{\text{out}}$ is the mean outbound screen weight; Δv_{cls} is the same-integrator classical asymptotic Δv (negative because of the asymptotic-shell truncation discussed in §5); HQIV–cls is the observable. The Anderson literature value is in the *Lit* column.

Mission-by-mission summary (nominal coupling).

Mission	Lit. [mm/s]	HQIV–cls [mm/s]	status	note
NEAR-1998 (1998-01-23)	+13.46	+16.34	matched	residual within drag systematics
Galileo I (1990-12-08)	+3.92	+4.15	matched	5% above literature
Cassini (1999-08-18)	-0.50	+0.74	excluded	thruster $\lesssim 1$ hr from CA
Rosetta-I (2005-03-04)	+1.80	-3.49	open	sign mismatch; above atmosphere

Ablation table. HQIV minus classical excess [mm/s] for the four catalog Earth flybys under the nominal coupling and three ablations. Numbers are direct CLI output from `scripts/hqiv_flyby_gap_report.py`.

Configuration	NEAR	Galileo I	Cassini	Rosetta-I
Lit. residual	+13.46	+3.92	−0.50	+1.80
Nominal (derived λ)	+16.34	+4.15	+0.74	−3.49
Fixed $\lambda = 0.5$	+33.46	+15.31	+9.80	−17.39
Drop L–T (isotropic only)	−1.91	+0.40	−0.56	+0.10
Drop colatitude + L–T	+48.71	+2.31	−1.00	+1.41
Drop modified geodesic $\mathbf{a}_{\text{GR}}/f$	+16.34	+4.15	+0.74	−3.49
Drop co-spin lapse ϵ_{spin}	~ 0	~ 0	~ 0	~ 0
Drop galactic disk Rindler	+16.34	+4.15	+0.74	−3.49

The picture is clear: the co-spin lapse ϵ_{spin} is the entire source of the HQIV excess at flyby accelerations; the derived $\lambda = \gamma \sin^2 \theta \rho_{\text{pol}}$ sets the magnitude and the sign of how that lapse is partitioned between an isotropic horizon force and a tangential Lense–Thirring force; the colatitude weight ensures the drag respects the spin-axis zero; and the modified-geodesic divide is dynamically subdominant at flyby altitudes ($w \sim 10^{-9}$). The galactic disk Rindler lapse contributes at 10^{-10} relative and is reported only for completeness.

Cassini exclusion. The -0.5 mm/s entry is reported but not scored. With thruster firings within roughly an hour of closest approach contributing an unmodeled Δv injection comparable to or larger than the reported residual, any HQIV value in the ~ 1 mm/s range is consistent with “the model is silent on this case”.

Rosetta-I open item. The sign mismatch persists across all repartition variants. Two hypotheses, both falsifiable:

- (1) Geometry: $\delta_{\text{in}} = -2.81^\circ$ is unusually small, so the inbound chord spends most of the close-approach time near the equator where $\sin^2 \theta \approx 1$ but $|L_z|/|L|$ is also large; the derived λ may be over-suppressing the L–T release in a regime where the literature data say the lapse must still flow. A precise SPICE/JPL ephemeris reconstruction is needed to confirm the impact-plane azimuth.
- (2) Physics: a low- $|L_z|$ sign reversal in the horizon channel that the current $\lambda = \gamma \sin^2 \theta \rho_{\text{pol}}$ does not capture. We do not introduce a new parameter to absorb this; the gap is reported honestly.

9. QUIET INTERSTELLAR VISITORS

The same integrator runs heliocentric hyperbolae with `--catalog interstellar: 1I/'Oumuamua, 2I/Borisov`, and generic ecliptic/polar templates. Central body is the Sun (φ readout at $R = 1$ AU, $m = 0$ horizon shell, no J_2 , no third-body tide), impact parameter from periapsis q and v_∞ , and early exit once the trajectory re-crosses the launch shell outbound. Non-gravitational acceleration (coma drag, outgassing) is not modelled. These runs are not scored against Anderson; they exercise the same equations of motion in a quiet, single-body, no- J_2 regime as a regression test against accidental Earth-centric specialisation in the codebase.

10. OPEN ITEMS

- **Lean orbit theorems.** Bundle the chart-level orbit hypotheses (`OMaxwellFluidChartHypothesis` style) so the derived λ , the horizon repartition, and the G_{eff} time-factor application are theorems rather than calculator conventions.
- **Asymptotic shell.** Replace the fixed $r_{\text{escape}} = 40 R_\oplus$ truncation by an adaptive condition $r \dot{v}_\infty / c < \epsilon$ to push absolute classical $|\Delta v|$ below mm/s without sacrificing wall-clock.

- **Per-mission ephemerides.** Replace the impact-plane azimuth fits with SPICE/JPL-derived ($\delta_{\text{in}}, \delta_{\text{out}}, \text{az}$) for each mission. This is the cleanest way to either resolve or sharpen the Rosetta-I sign mismatch.
- $\kappa_{\text{vac}}, \kappa_{\varphi}$ **from first principles.** Map the two chart-coupling scales to a non-fitted SI normalisation derived from the lattice ladder, so the paper’s nominal coupling is itself a Lean theorem rather than a chosen value.
- **NEAR atmospheric drag posterior.** Combine the published drag uncertainty band with the HQIV residual (+13.46 literature, +16.34 model) to produce a posterior on the residual “unexplained” anomaly. The current gap of ~ 3 mm/s is within the drag systematic.

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`scripts/hqiv_orbital_flyby_omaxwell.py`, `scripts/hqiv_flyby_equations.py`,
`scripts/hqiv_flyby_gap_report.py`.

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